Submitted to:
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Robotic Navigator Challenge Proposal

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I. Abstract

Autonomous robots are the way of the future. Currently autonomous robots are used in a variety of areas such as NASA, industrial factories and even in households. The task at hand is to design, build and implement an autonomous robot that can navigate a course set by five beacons in the shortest possible time. The Power Robotics design will consist of a tank base along with the use of ultrasonic sensors, object avoidance and wall following algorithms to navigate through the course.

II. Project Description

The design requirements for the project are that the robot navigates through a predetermined course completely autonomously. There are size constraints limiting the dimensions of the robot to 16”x16”x16”. The last requirement is that there must be an emergency stop switch.

The project, as a whole, is divided into two sectors: navigation and locomotion. The navigation team is responsible for navigational sensors and the microcontroller. First, the robots central processor, the PIC32, will rate the inputs from the sensors based on which section of the course it’s on. The highest rated input will take priority and based on the input it receives will determine the direction of the robot. The team will program ultrasonic sensors for object avoidance and beacon detections. When distance travelled and turning radius is needed the dedicated microcontroller will use input from the Hall Effect sensor for accuracy.

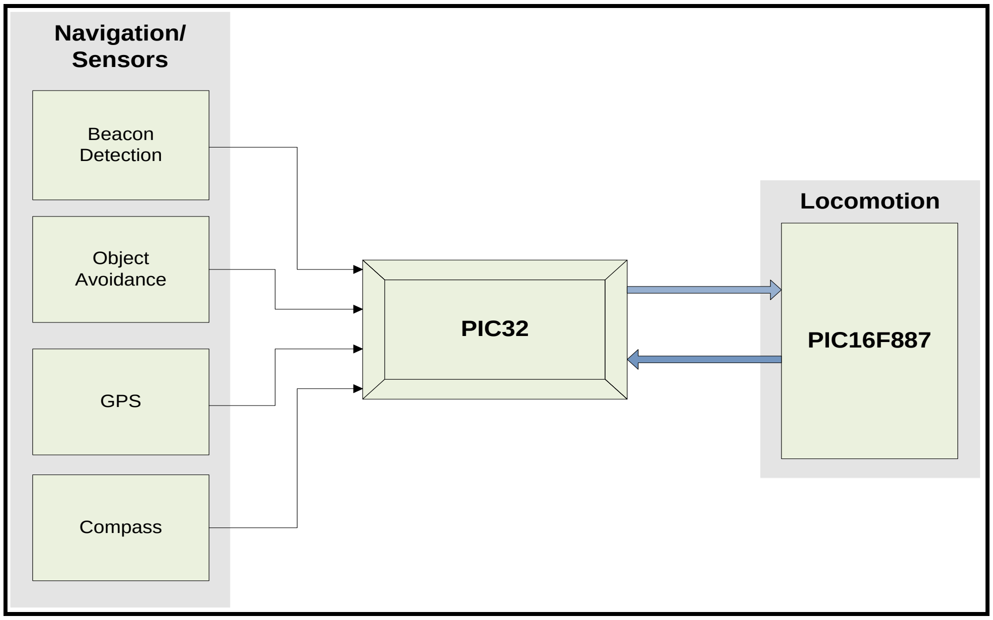
The locomotion team is responsible for the base design, power supply and motor controllers. The Power Robotics team decided to implement a tank based design due to the rugged terrain along the course. The tank’s treads will be able to easily maneuver through the hills, potholes, and other uneven surfaces. The main advantage of using the tank design will be at the end of the course, where the Power Robotics team took the option to go down the stairs instead of down the handicapped ramp. Two 7.2VDC batteries will be connected in series to the power supply to power the motors through the h-bridge. The second part of the power supply will have an input of 6VDC and output 3.3V to the compass, and 5V to the GPS, navigation sensors and microcontroller.

III. System Design

Navigation:

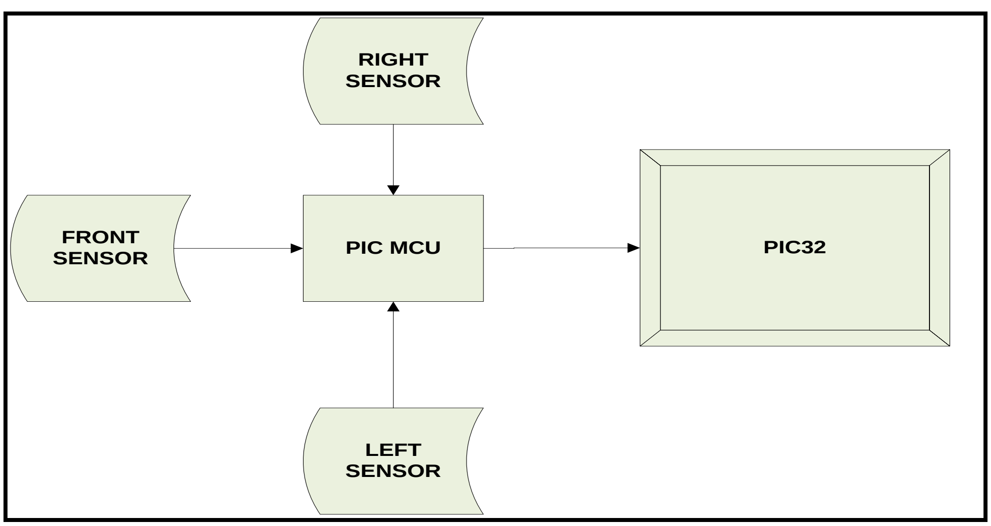
Microcontroller

The PIC32MX460F512L will be the main microcontroller that will take in the different data from the sensors and choose the proper navigational algorithm to send out to the H-bridge microcontroller to move the robot in the correct direction. The PIC32 will make all decisions on where the robot needs to go. Specifically, the GPS and compass will be directly controlled by the PIC32. The remaining sensors, which include the ultrasonic beacon, motion detection, and Hall Effect sensors, will be controlled by their own dedicated microcontrollers that will interpret the immediate outside data and relay it back to the PIC32. Even though these other sensors have their own microcontrollers, the CerebotMX4 will decide how to organize and prioritize all the robots data to make the correct decisions.



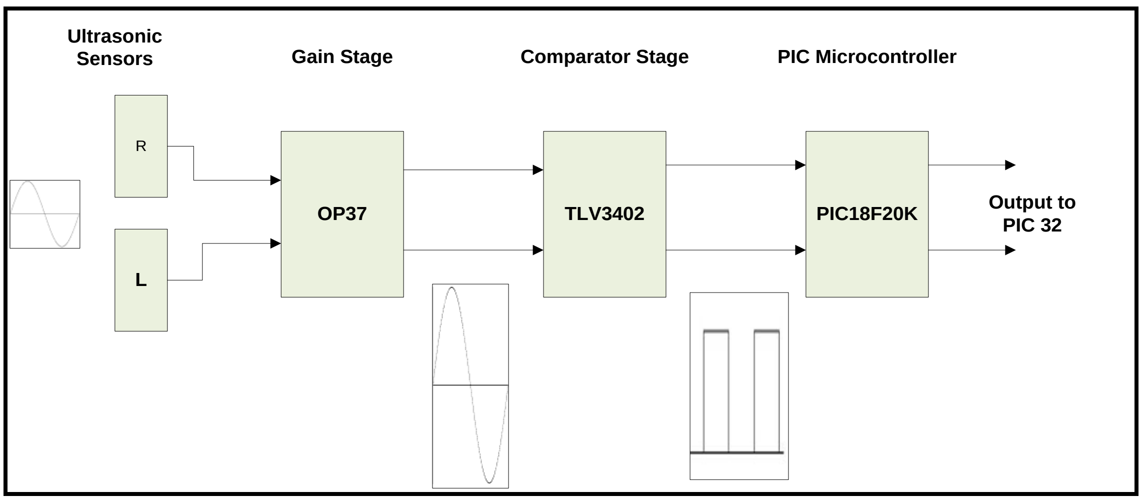
Object Avoidance

While navigating through the course, objects along the way might be blocking the robot. In order to avoid such objects and move successfully around, Power Robotics will equip the robot with ultrasonic range finder sensors. There will be a total of 3 sensors, a front sensor, a left-side sensor and a right-side sensor connecting to the robot. These sensors will detect objects that are 5 ft in its forward path and 8 inches detecting objects on the side. The sensors will help the robot move around objects safely in its path and or perform wall hugging guidance. In order to communicate, the sensors will be connected with a PIC microcontroller, outputting the available options to move after encountering onward obstacles to the main microcontroller.



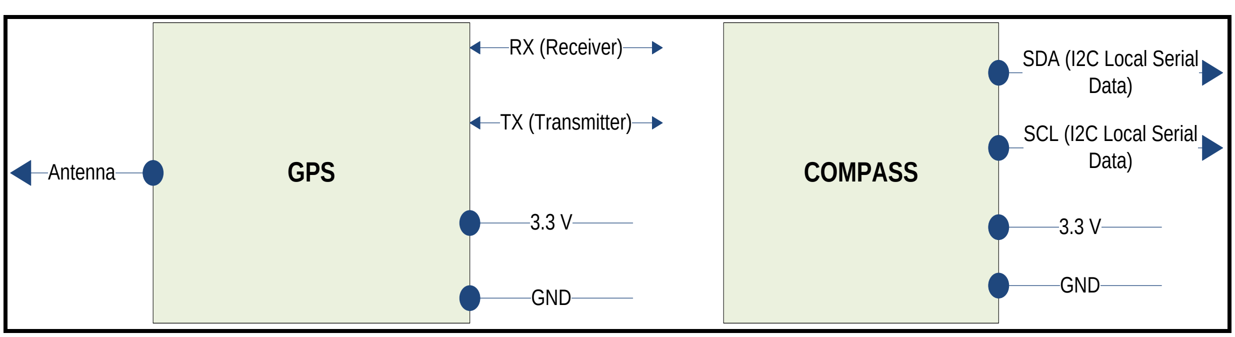
Beacon Detection

Throughout the course there will be 5 gates each equipped with a 25 kHz ultrasonic beacon which sends out a signal for the robot to track and follow. Power Robotics will use two ultrasonic sensors spaced approximately one foot apart. The spacing between the sensors will allow the robot to triangulate the beacon by sensing the time delay between when each sensor detects the beacon. The process begins by sensing the signal and feeding it into the Gain Stage which consists of a low noise, high gain amplifier. The amplifier will amplify the signal by approximately 100v/v. This amplified signal will then be fed into the Comparator Stage which will change the sine wave into a digital signal of either Low (0V) or High (5V). This digital signal is then fed into a PIC microcontroller which will determine through an algorithm to which sensor detected the beacon first. The PIC will then send out directional data to the PIC 32 that tells which direction the beacon is with respect to the robot.



GPS & Compass

The GPS will tell the microcontroller the position of the robot. There are five wires connected to the GPS, four of them go to the microcontroller and one connected to the Antenna. The GPS consists of two parts: satellites orbiting the Earth and control/monitoring stations on the Earth. GPS receivers pick up the signals from space and identify them. Each GPS receiver then provides three-dimensional location (latitude, longitude, and altitude) plus the time. The compass will be used to feed the robots current directional heading into the PIC 32.



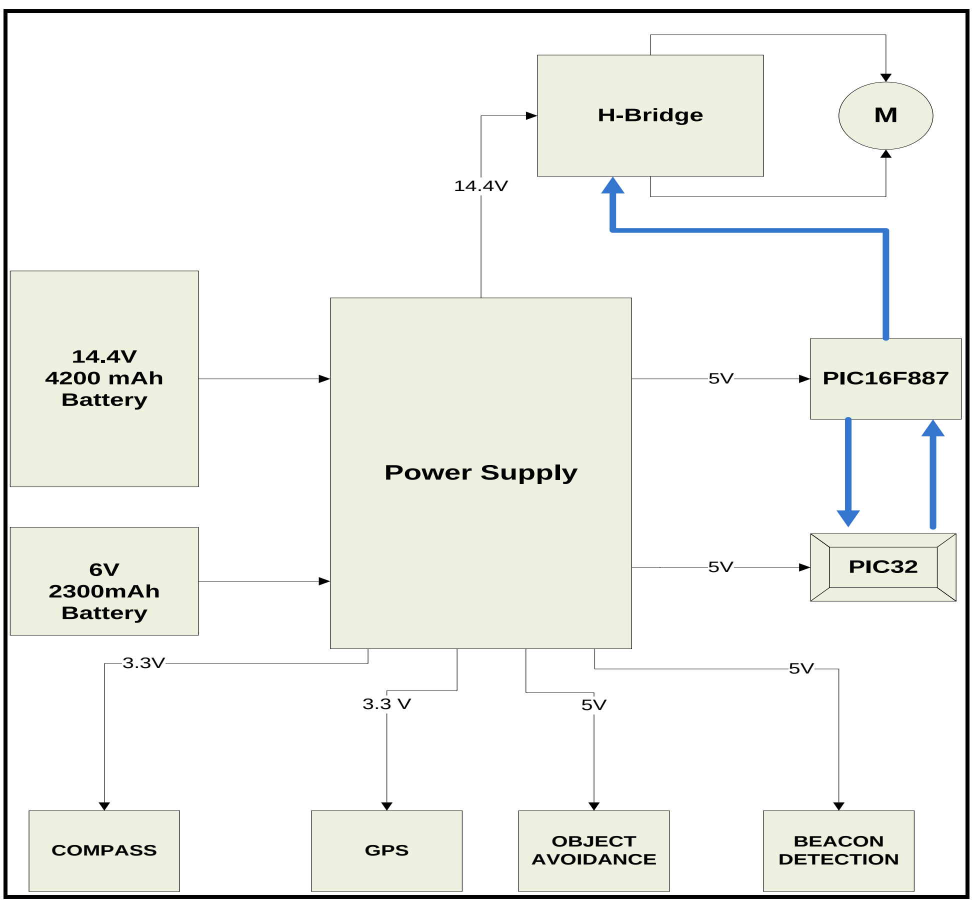
Locomotion:

Power Supply

Power will be supplied by two 7.2 VDC batteries and one 6 VDC battery. These batteries will be connected to a power supply board that will distribute the right amount of voltage output to the entire robot's electrical design, which includes the navigation electronics and locomotion/control systems. The navigation electronics, which includes the PIC32, PIC16F887, sensors, and compass, are powered by lower voltages, therefore multiple outputs with a clean and noise free 3.3 V and 5 V will be built. The locomotion/control systems, which include the H-bridge and motors, will be powered with a slightly higher voltage of 14.4 V. The power supply board is responsible for distributing clean power flow, overall system power organization and system protection.

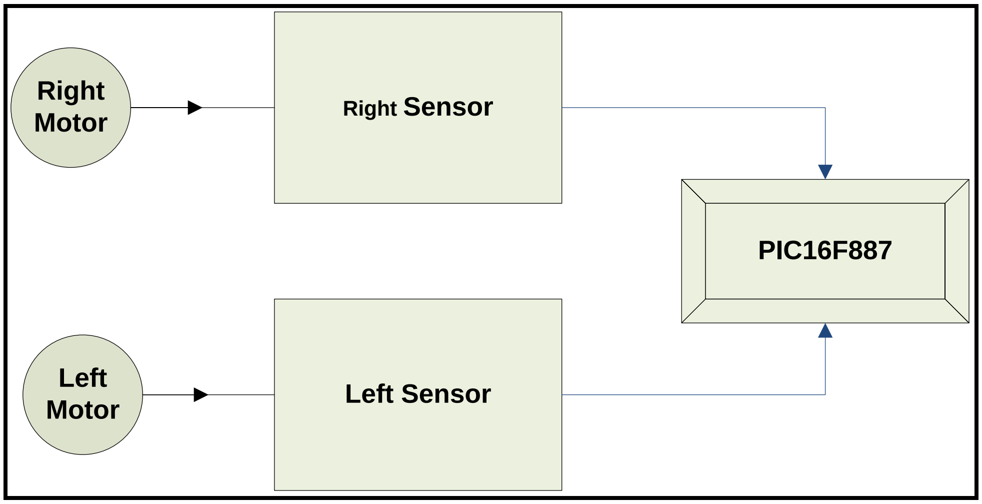
Wheel Encoding

The PIC16F887 will be used to encode the tracks for the robot to run smoothly through the course. In order to maintain the robots progress in a straight line, the wheels must be at the same speed. The ability to turn at the right speed is also critical. The PIC16F887 will be programmed using MPlab software in assembly language to accomplish these tasks by outputting certain amounts of duty cycles to the H-bridge which controls the motors.

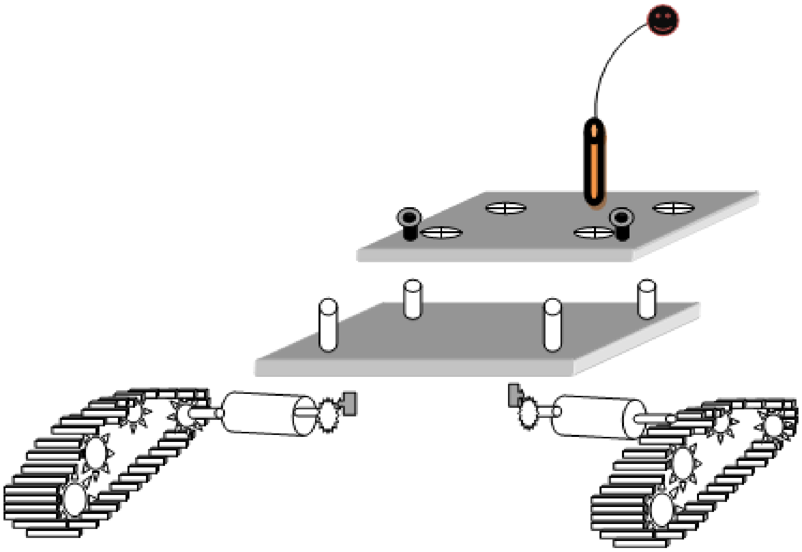


HALL EFFECT SENSOR

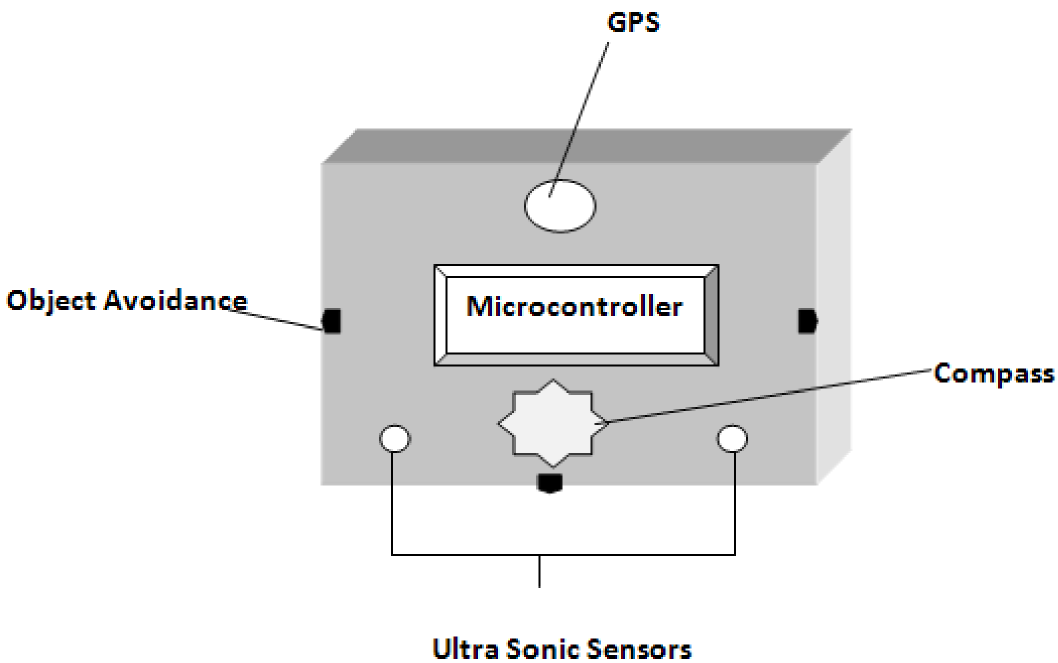
The Hall Effect sensors responsibility is to measure the distance and speed travelled by the robot. Its task is to sense the magnetic flux of the robots left and right motor shafts by placing minute magnets on a sprocket attached to the shaft. Detecting this flux fluctuation the sensor will then produce a high (1) or a low (0). This information will be converted into data in the form of miles per hour and distance travelled. This will then be connected to the PIC16F887 where the data will be compared and will either speed up the motors or slow down the motors.



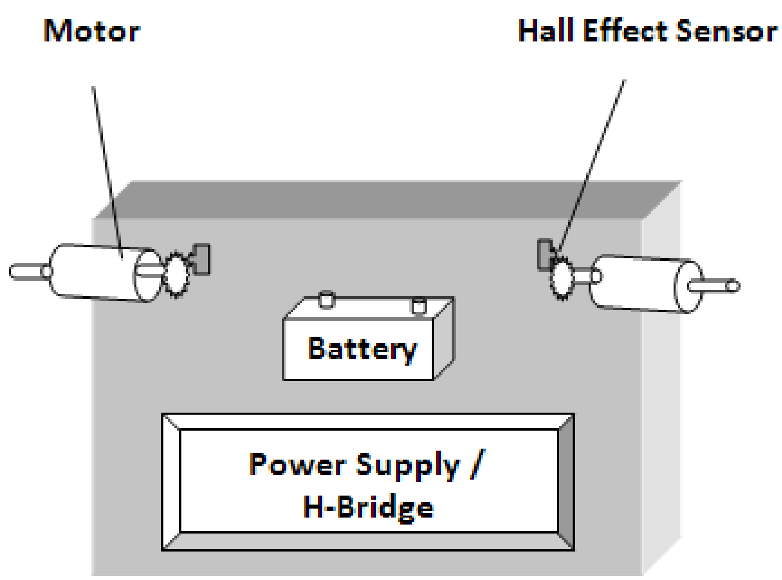
TANK LAYOUT DESIGN



Top View Upper Level



Top View Lower Level



IV. ***Project plan***

Constraints

The economic constraint for the project is $750 for all parts. The robot must also be no larger than 16”x16”x16”. The next design constraint is the rugged terrain that the robot will be maneuvering through. The terrain is made up of hills, holes, turns and other miscellaneous objects and the robot must be able to withstand navigating its way through the course numerous times. At the end of the course, there is the option to have the robot navigate its way down the wheelchair ramp or the stairs, the Power Robotics team opted to build a tank design capable of traversing the stairs.

Resources

Each team member will be assigned a specific part to research and specialize in. The tasks were split up as follows:

Navigation:

Ben Lauer – Beacon Detection/Project Manager
Nick Lerch – Microcontroller Programmer/Parts Manger
Christopher Abelon – Object Avoidance/Presentation Editor
Christopher Lamorena – Microcontroller Programmer/Website Designer
Baqer Al-Shakhs – GPS & Compass/Website Designer

Locomotion:

Nicole Bavard – H-bridge circuit Design/Project Manager
Michael Chen – Sensor & Base Design/Parts & Budget Manager
David Beltran – Sensor & Base Design/Technical Illustrator
Alex De Leon – Power Supply Design/PIC16F887/Report Editor
Ariel Cuaresma – Power Supply Design/ PIC16F887/Presentation Editor

Cost Analysis

Navigation:	
PIC32Mx4 Cerebot	$60.00
Pmod Switch	$10.00
Ultrasonic Transducers	$2.00
GPS	$50.00
Antenna	$12.00
Ultrasonic Sensors	$75.00
Miscellaneous	$75.00
	$466

Locomotion:	
Tank Base/Tracks	$286
Batteries	$60.00
H-Bridge	$20.00
Sensors	$20.00
Miscellaneous	$80.00
	$284

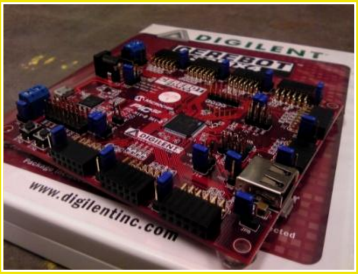
Total **$750**

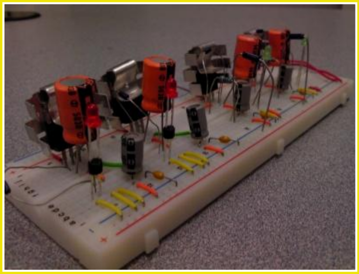
Project Plan (Gantt Chart)

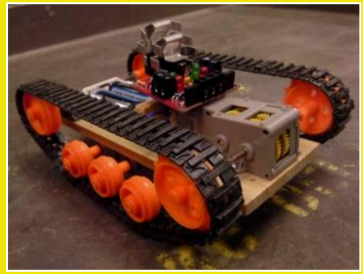
ID	Task Name	Duration	Start	Finish	Predecessors	Resource Names
1	**Navigation**	**50 days**	**Sat 2/27/10**	**Mon 5/3/10**		
2	PIC32 Microcontroller Programing	50 days	Sat 2/27/10	Mon 5/3/10		Chris / Nick
3	**Ultrasonic Beacon Detection**	**19 days**	**Sat 2/27/10**	**Mon 3/22/10**		**Ben**
4	US Beacon Detection Circuit	7 days	Sat 2/27/10	Mon 3/8/10		
5	US Beacon Detection PIC programir	12 days	Tue 3/9/10	Mon 3/22/10	4	
6	US PIC Microcontroller	7 days	Tue 3/9/10	Tue 3/16/10	4	
7	**Object Avoidance**	**18 days**	**Thu 3/4/10**	**Thu 3/25/10**		**Chris**
8	Research	6 days	Thu 3/4/10	Thu 3/11/10		
9	Test and Program	12 days	Fri 3/12/10	Thu 3/25/10	8	
10	**Location**	**12 days**	**Sat 2/27/10**	**Sat 3/13/10**		**Baquer**
11	**GPS**	**6 days**	**Sat 2/27/10**	**Fri 3/5/10**		
12	Test and Program	6 days	Sat 2/27/10	Fri 3/5/10		
13	**Compass**	**6 days**	**Mon 3/8/10**	**Sat 3/13/10**	**11**	
14	Test and Program	6 days	Mon 3/8/10	Sat 3/13/10		
15						
16	**Locomotion Team**	**51 days**	**Mon 3/1/10**	**Wed 5/5/10**		
17	Project Management	50 days	Tue 3/2/10	Wed 5/5/10		
18	**Base**	**12 days**	**Tue 3/2/10**	**Tue 3/16/10**		**Mike**
19	Construct	10 days	Tue 3/2/10	Sat 3/13/10		
20	BASE COMPLETE	3 days	Sat 3/13/10	Tue 3/16/10		
21	**Power Supply**	**15 days**	**Mon 3/1/10**	**Thu 3/18/10**		**Alex/Ariel**
22	Test	4 days	Mon 3/1/10	Thu 3/4/10		
23	Impelment	10 days	Mon 3/8/10	Thu 3/18/10	22	
24	**Batteries**	**6 days**	**Fri 3/19/10**	**Thu 3/25/10**	**21**	**Alex**
25	Test	2 days	Fri 3/19/10	Sat 3/20/10		
26	Implement	4 days	Mon 3/22/10	Thu 3/25/10	25	
27	**H-Bridge**	**8 days**	**Fri 3/19/10**	**Sat 3/27/10**	**21**	**Nicole**
28	Test	2 days	Fri 3/19/10	Sat 3/20/10		
29	Implement on board	6 days	Mon 3/22/10	Sat 3/27/10	28	
30	Programming PIC PWM	0 days	Sat 3/27/10	Sat 3/27/10		
31	**Hall Effect Sensors**	**7 days**	**Fri 3/19/10**	**Fri 3/26/10**	**21**	**David**
32	Test	2 days	Fri 3/19/10	Sat 3/20/10		
33	Implement	5 days	Mon 3/22/10	Fri 3/26/10	32	
34						
35	**Combine Navigation & Locomotion**	**29 days**	**Mon 3/29/10**	**Thu 5/6/10**		
36	Test batt/PS/H-bridge together	3 days	Mon 3/29/10	Wed 3/31/10	27,24,21	
37	Add batt/PS/H-bridge to base	2 days	Thu 4/1/10	Fri 4/2/10	36	
38	Use Beacon Detection Code to locomotic	3 days	Mon 4/5/10	Wed 4/7/10	37,3	
39	Use Object Avoidance code with locomo	3 days	Thu 4/8/10	Mon 4/12/10	38,7	
40	Add GPS/Compass to locomotion	6 days	Tue 4/13/10	Tue 4/20/10	11,13,39	
41	Test completed Robot	0 days	Tue 4/20/10	Tue 4/20/10	40	
42	Debug Robot	13 days	Tue 4/20/10	Thu 5/6/10		

POWER ROBOTICS

Advancing Technology







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